Akarsh Dubey

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.12
ST MARY'S SCHOOL	Class XII (ISC)	2019	87.40%
ST MARY'S SCHOOL	Class X (ICSE)	2017	87.17%

THESIS

Rehearsal-Free Prompt Federated Continual Learning (Ongoing) || Guide: Prof C. Krishna Mohan. ||**March 2026**

- Investigating prompt-based federated continual learning (FCL) to mitigate catastrophic forgetting across distributed medical clients without storing or sharing any patient data (rehearsal-free).
- Adapting Fed-CPrompt (ICML 2023) to medical imaging: task-specific prompts learned on a frozen pre-trained ViT via asynchronous prompt aggregation across clients with heterogeneous data distributions.
- Introducing a Contrastive Continual Loss (C2Loss) to enforce distinct task-specific prompt construction and reduce inter-client drift, targeting class-incremental FCL across sequential medical imaging tasks.
- Diagnosed catastrophic forgetting in the trainable federated classifier through task-wise accuracy and forgetting analysis, showing severe recency bias and performance degradation on non-IID medical clients.

PROJECTS

Image Restoration -- KLA x IIT-H 2026 Hackathon || HAT-L Transformer | PyTorch || **Jan 2026**

- Built an end-to-end joint denoising and 2× super-resolution pipeline for degraded grayscale images, reconstructing 256×256 outputs from low-resolution inputs affected by speckle noise and blur.
- Adapted the HAT-L (Hybrid Attention Transformer) with ImageNet SR×2 pretrained weights and a Charbonnier + MS-SSIM + FFT-L1 composite objective, along with synthetic degradation and Gamma speckle augmentation.
- Enhanced restoration robustness with EMA inference, multi-scale overlap-tile inference, and 8-fold D4 TTA.
- Implemented multi-scale overlap-tile inference and restoration post-processing to reduce boundary artifacts and improve reconstruction quality on degraded high-resolution images.

SmartRAG -- Adaptive Agentic RAG Assistant || DistilBERT, FAISS, Sentence-Transformers, Streamlit ||**Aug 2026**

- Built an adaptive RAG pipeline over a FAISS index of 47 Wikipedia networking articles, supporting direct answering, single-step retrieval, and multi-step agentic search with citation-grounded responses.
- Developed a query-complexity router using DistilBERT to select the appropriate answering strategy, with lightweight fallback models and automatic escalation when single-step retrieval fails to provide sufficient evidence.
- Implemented iterative agentic retrieval and evaluation tooling, enabling the system to reformulate queries, perform up to three searches, track retrieval/LLM cost, and compare adaptive routing against fixed retrieval strategies.
- Added citation grounding, retrieval-failure detection, and cost/step tracking to evaluate answer reliability while minimizing unnecessary multi-step agentic searches.

Multilingual Business Entity Resolution | Amazon ML Challenge 2026 || Python, PyTorch, LightGBM, Transformers || **Sept 2026**

- Engineered multilingual retrieval pipeline that matches noisy business records across 3 sources with no shared identifiers. It combines character 3-gram TF-IDF, exact-key blocking and a fine-tuned multilingual-e5-small kNN search, with Indic-script transliteration.
- Fine-tuned 3 cross-encoders (XLM-RoBERTa-large, mDeBERTa-v3, BGE-reranker-v2-m3) on held-out data. Their scores go into a two-stage LightGBM ranker with context features (per-record rank, competition between records, agreement between related candidates).
- Designed a decision layer that applies a one-to-one assignment and then picks the match set that maximises expected F0.5. A LightGBM pre-filter cuts the candidates to about 5 per record, which keeps inference fast and precise.
- Generalized to France, a country absent from the training data, by fine-tuning on synthetic French pairs made from labelled US data. This beat self-training and proxy validation. **The final score was 0.98964 F0.5, ranking 63rd (top 0.2%) among 32,000+ teams.**

CPU Scheduling Simulator || C++, Node.js, Express, JavaScript. || **Jan 2026**

- Engineered an Operating Systems-based CPU scheduling simulator implementing 10 scheduling algorithms including FCFS, SJF, SRTF, Priority, Round Robin, HRRN, LJF, LRTF, Multilevel Queue, and MLFQ.
- Implemented preemptive and non-preemptive scheduling algorithms using FIFO queues, priority queues, and process-state management, computing completion time, waiting time, turnaround time, and CPU execution traces.
- Integrated the C++ scheduling engine with a Node.js/Express.js RESTful API using JSON-based communication, enabling configurable process workloads and seamless backend-frontend interaction.
- Developed an interactive HTML/CSS/JavaScript interface with Gantt-chart visualization and scheduling metrics, and added automated functional testing with manually verified reference cases for FCFS, SJF, and Round Robin.

SKILLS

Programming Languages: C, C++, JavaScript, Python

Web Tech & Database: HTML, CSS, React.js, SQL

Machine Learning: Numpy, Pandas, Sklearn, PyTorch, Matplotlib, Seaborn, Hugging Face

Deep Learning: CNN, RNN, LSTM, GRU, Transformer, Autoencoder

Tools: Wireshark, tcpdump,GitHub, Jupyter, dnspython, pycryptodome

RELEVANT COURSES

Foundations of Machine Learning || Deep Learning || Fraud Analytics || Advanced Computer Networks || Advanced Data Structures and Algorithms || Network Security || Visual Computing

POSITIONS OF RESPONSIBILITY

Placement Coordinator OCS

- Coordinated placement activities and supported student-recruiter interactions.

Coordinator, Japan Career Day | IIT Hyderabad

- Coordinated Japan Career Day at IIT Hyderabad, managing recruiter-student communication, event logistics, scheduling, registrations, and on-ground execution for a campus-wide career networking event.